

# Engagement With a Coordination Game Decoupled From Fall Risk Assessment: Exploratory Observational Study

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## Abstract

**Background:** Failure-induced abandonment syndrome (FIAS) has been proposed to describe withdrawal after a personally salient adverse experience with gamified fall risk assessment (FRA). A candidate explanation concerns appraisal of one's own capability. Removing clinical apparatus and outputs offers a way to investigate this question, although changes in game design, access and social context may also affect participation.

**Objective:** To describe engagement in a coordination game decoupled from FRA apparatus and outputs, and examine its implications for investigating FIAS and learning from gameplay experience.

**Methods:** Stage 3A was an exploratory observational deployment in March–April 2026. Twenty distributors were invited, each asked to recruit 10 players across age groups. Actual invitations were unknown; the summary contained 120 returned gameplay records from adults aged 25–80 years, all new to this study game. Delivery included emailed instructions and prepared iPads, with foot-pad and/or keypad controls. Aggregate engagement classifications were analysed descriptively. Clinical performance, self-appraisal and learning were not assessed in Stage 3A. Subsequent Stage 3B Pro development exports were examined separately as a developmental record.

**Results:** The Stage 3A summary classified 23 of 120 records (19.2%) as initial dropouts and 15 of these 23 (65.2%) as returned. Initial-dropout classifications were 0 of 38 (0.0%), 7 of 41 (17.1%) and 16 of 41 (39.0%) in the 25–44, 45–64 and 65–80 age bands. Unavailable applied classification rules and event timing limit interpretation.

**Conclusions:** The deployment supports a research direction in which abandonment is investigated in relation to self-appraisal and opportunities to learn from experience. It does not establish FIAS as a distinct mechanism, an effect of clinical decoupling, or adaptive-learning efficacy. Source-linked behavioural records and direct measures of appraisal and learning are needed to test these propositions.

**Keywords:** gamification; older adults; engagement; self-appraisal; coordination game; fall risk assessment; exploratory observational study; adaptive learning

# Introduction

## Background

A survey of 200 residents at a Guangzhou care facility, from a resident population of 280, found that 39% had used an installed fall risk assessment (FRA) system; among prior users, approximately 74% reported an adverse or unacceptable first experience and, of that subgroup, approximately 82% reported no subsequent return [1]. A subsequent eight-week deployment with 30 residents examined engagement with a gamified FRA at the clinical station. In that cohort, 70.0% recorded three or fewer completed runs and none of 21 low-engagement participants recorded a later run during the observation window [2]. These earlier observations motivated a redesign of the way participants encountered the game.

The programme proposed failure-induced abandonment syndrome (FIAS) as a provisional label for non-return following an adverse, personally salient experience [1,2]. The central hypothesis concerns a self-cognitive challenge: a player may interpret performance in a clinically attached game as evidence about their own capability, making further participation personally threatening. FIAS is not a clinical diagnosis or a validated measurement construct. Technology-acceptance and gamification approaches remain relevant [3,4,5]; the senior technology acceptance model already includes self-efficacy and anxiety [4]. The present hypothesis therefore requires evidence of a specific appraisal process, not simply evidence of abandonment. Loss aversion and the endowment effect offer possible theoretical analogies [6,7,8], but neither these processes nor identity threat were measured here.

Gerontoludic design prioritises enjoyable play and the diversity of ageing [9], while affective game planning explicitly connects appraisal to meaningful interaction [10]. These perspectives support examining the meaning a player assigns to an experience. Usability, accessibility, motivation and ordinary attrition also remain possible explanations for non-return [11]. Clinical decoupling is a design strategy for investigating the appraisal hypothesis; it cannot by itself verify FIAS or exclude these alternatives.

## Study Objectives

Stage 3A transferred the skiing-style coordination game to a browser deployment without the FRA assessment apparatus or clinical outputs. Its primary objective was to describe the available engagement classifications and their age distribution in this setting. Its conceptual objective was to examine how this decoupling could inform investigation of FIAS as a proposed self-appraisal mechanism. A further design objective concerned learning from experience: continued play could provide opportunities to adjust strategies and responses. Stage 3A did not measure such learning. Subsequent Stage 3B development explored alternative mechanisms for a future adaptive-learning evaluation and is discussed separately from the Stage 3A results.

## Methods

### Study Design and Setting

Stage 3A was an exploratory observational deployment conducted in March–April 2026, reported through aggregate tabulations of engagement classifications. Multimedia Appendix 1 maps the report to the STROBE checklist for cross-sectional studies [12] and identifies unavailable information. Recruitment used Hong Kong-based personal and community networks, including work contacts, a charity community network, university support channels and medical-centre peers. Gameplay

occurred away from the clinical FRA station. Supervision was not required. Some older players received help operating a prepared iPad from a research assistant, without the assistant playing on their behalf. The number and extent of assisted sessions were not recorded by participant. The therapist-supervised Stage 2 cohort [2] provides historical context and was not a concurrent control group.

## **Participants and Recruitment**

We invited 20 distributors, acting as research assistants, and asked each to find 10 players across age groups, giving a nominal recruitment target of 200. The number of distributors who actively recruited, the number of people they actually invited and the number who began play without returning a file were not compiled. The target was a recruitment aim rather than a formal power calculation. We received 120 gameplay datasets represented in the aggregate summary. Accordingly, 120 is the returned-record denominator; it is not an invitation count, and dividing it by the target of 200 would not provide a response rate.

Stage 3A players had not previously played this study game. This refers to study-specific familiarity, not general gaming or technology experience. Stage 3A used a different cohort from the Stage 2 residents. The summary records ages of 25–80 years: 38 records in the 25–44 band, 41 in the 45–64 band and 41 in the 65–80 band; 48 were recorded as men and 72 as women, and all as having high-school education or above. These demographic counts are retained as compiled. Distributor briefs addressed players aged 18–45 and players aged 45 and over; these brief categories differ from the reported analysis bands. Recruitment through distributors and onward forwarding constitutes a nonprobability network sample with possible selection and researcher influence. Participant-level linkage and duplicate checking are required to confirm the correspondence between returned files and unique participants.

## **Participant Information and Game Presentation**

Distribution included remote email invitations with introductory information and prepared iPads with the game preloaded for some older players. We describe the participant invitation as a university-headed invitation to play a skiing game, introducing falls and balance training and disclosing the game's origin in an FRA platform. The available invitation document is a retrospective account dated 4 August 2026; the original version as distributed has not been linked to the study record. A distributor email described a coordination game and stated that the clinical context was being dropped. The original communications are needed to establish the health-related wording each participant received.

Clinical decoupling removed the pressure-panel assessment station, the pre-session clinical scan and participant-facing clinical scores or risk outputs. Stage 3A used foot-pad and/or keypad controls, intended to approximate aspects of the control demands of Stage 2. This intended similarity does not establish equivalent motor demands or measurement validity. Device, location and assistance could vary, and recruitment could still mention the health-related origin. Decoupling therefore denotes removal of clinical apparatus and outputs within a broader deployment change; exposure to health-related framing was not experimentally controlled.

## **Game Build and Intended Procedure**

The archived Stage 3A build is `Alpine_Coordination_Game_Stage3_v2.2.html`, a 37,279-byte HTML file with SHA-256 prefix 42bc3df6; the archive identifies it as byte-identical to the `_v2` file. Runs use

a distance goal or a time cap: 200 m or 30 seconds, and 400 m or 60 seconds. The zero-failure mode does not slow the player following a bump. The build supports keyboard arrows and a joystick or foot pad through the browser Gamepad API; it lacks touch controls. Prepared-iPad delivery therefore does not imply touchscreen input: the reported tablet-assisted sessions used an external keyboard. Participant-specific device and controller combinations are unavailable. Touchscreen input belongs to the later Stage 3B design. Figure 1 illustrates the archived Stage 3A design at staged difficulty settings.

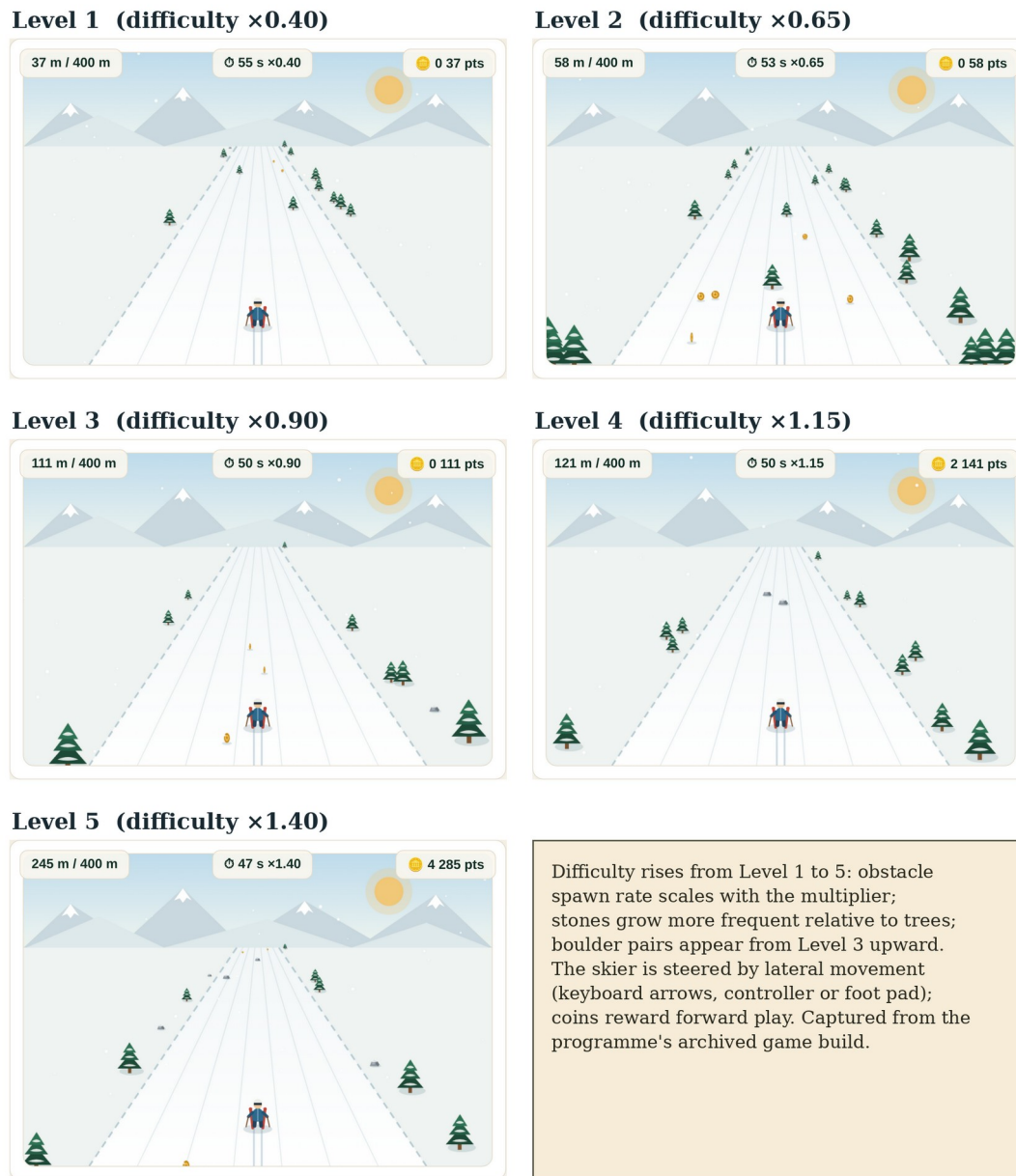


Figure 1. Archived Stage 3A coordination-game design at staged difficulty multipliers. Steering uses keyboard arrows, a controller or foot pad. The interface omits clinical labels, clinical scores and risk outputs. These screenshots illustrate the archived design and do not verify the settings or version used by every participant.

The distributor brief suggested five sessions, whereas the aggregate register describes initial dropout as stopping before five attempts. Whether these units meant separate visits or runs within one sitting is not established. The intended procedure is therefore distinguished from verified completion. The

build includes an enabled-by-default option to request language-model difficulty advice, with a local-rule fallback. The absence of an embedded credential and the browser configuration suggest that local rules operated, but no retained runtime log establishes whether a model request succeeded. AI assistance in software development is separate from an AI system operating during gameplay; no language-model effectiveness is claimed.

## **Outcomes and Source Records**

The primary available quantities are the register's classifications of initial dropout, returned among initial dropouts, and non-return among initial dropouts. The descriptive initial-dropout label is "dropped out before completing 5 attempts"; the rule actually applied in aggregation is not retained. Dates of first play, disengagement and return, a disengagement checkpoint, reminders and assistance are unavailable in the compiled record. Consequently, these counts cannot distinguish immediate replay from later re-engagement or prompted from unprompted return. The complement of the eight non-return classifications is 112 records, which does not establish five-attempt completion or activity at an observation-window endpoint and is not a retention outcome.

Two questionnaire-related counts are retained as positive-coded responses in the aggregate summary. The summary labels the items as liking the game as a reflection of eye-leg coordination and using it for regular health training if accessible; the retrospective invitation instead describes a coordination challenge and willingness to play regularly. The administered wording, language, response scale and positivity threshold have not been reconciled. These counts are not treated as validated measures of coordination, intrinsic motivation, intention or the success of a framing manipulation.

All Stage 3A numerical results reported here originate in the Preliminary\_Summary sheet of the Stage 3 data logbook. Its participant-level sheets do not yet contain the source-linked observations needed to regenerate the summary or verify unique participant counts. Returned exports and research records are retained, but record-level exclusions, duplicate checks and missingness cannot be reconstructed from the summary. No imputation or participant-level adjustment was performed. Records from subsequent development exercises are not added to the Stage 3A denominator.

## **Statistical Analysis**

Counts and percentages are reported with explicit denominators. Wilson score 95% confidence intervals use the reported binomial counts. An exploratory chi-square test examines age-band differences in initial-dropout classifications, and a Fisher exact test compares the youngest and oldest bands. These unadjusted comparisons were selected after tabulation; they were not prespecified and no multiplicity adjustment was applied. Analyses use the available aggregate age bands, without participant-level adjustment for device, assistance, education, prior technology experience or distributor clustering. Intervals and tests are conditional on the reported counts and assumed independence; they do not resolve selection or classification uncertainty. No causal contrast with Stage 2, time-to-event analysis, paired behaviour-intention analysis, sensitivity analysis or learning-effect estimate was calculated.

## **Stage 3B Pro Development Records**

Eight Pro exports from 19–20 September 2026 (UTC; builds 4.1 and 4.1.1) were examined retrospectively as a separate development exercise. The investigator confirmed that their four codes were assigned to four distinct volunteers who also invited other players. The exports do not enumerate those invitees or establish whether any used an existing volunteer code. Counts therefore

refer to codes, runs and software-defined visits, without assuming verified individual attribution of every run or independent participants for each visit.

Repeated cumulative exports were reconciled by code and run identifier; duplicate practice-detail rows in CSV exports were not counted again. Adaptation decisions were checked against linked performance and feedback records and the next run's recorded settings. Complete within-visit opening and closing checks were described using obstacle clearance, calculated as  $100 \times (\text{obstacles faced} - \text{obstacles hit}) / \text{obstacles faced}$ . These descriptive checks do not estimate a learning effect or establish equivalence of different courses.

## **Ethical Considerations**

Participation was voluntary and unpaid. According to the retained account of the invitation and distribution emails, participants were informed about the study, collection of game records and questionnaire responses, aggregate use, voluntariness and withdrawal. Returning the results file and questionnaire responses constituted implied informed consent; no written consent instrument was administered. No personal names were requested at enrolment. Analysis identifiers are separated from identifying correspondence. Stage 3A did not conduct a clinical health-performance assessment.

Stage 3A collection occurred in March–April 2026 after the supervisor's confirmation to proceed. An administrative misunderstanding between the researcher and the administration office delayed the formal application; this account has not been linked to a contemporaneous authorisation record. The programme's human-subjects ethics application, HSEARS20260805008, was submitted and approved on 6 August 2026 by the Departmental Research Committee of the Department of Industrial and Systems Engineering on behalf of The Hong Kong Polytechnic University Institutional Review Board, with a stated approval period of 16 June 2024 to 30 October 2026. The application described Stages 1–3 as already conducted. Thus, the formal decision followed Stage 3A collection. The supervisor's confirmation and the later formal decision are distinct. Institutional clarification of the applicable authorisation and the decision's treatment of the earlier procedures remains outstanding.

The Pro exports retain the software label “researcher-self-test”, with participant-release or approval fields set to false and ethics references absent or not asserted. The investigator's confirmation of volunteer use is reported separately from those settings. The applicable consent and institutional-authorisation records for this use require reconciliation; the software flags alone do not establish its ethics status.

## **Results**

### **Compiled Classification Counts**

Of the 120 records in the summary, 23 (19.2%) carried an initial-dropout classification. Fifteen of those 23 (65.2%; 95% CI 44.9%–81.2%) also carried a returned classification and eight (6.7% of all records) a non-return classification. The remaining 97 records are the arithmetic complement of the initial-dropout count. Table 1 presents these quantities without treating the labels as a verified sequence of behavioural events.

Table 1. Compiled Stage 3A classifications with explicit denominators and conditional Wilson 95% confidence intervals.

| Classification                          | Count / denominator | Percentage | 95% CI in %  |
|-----------------------------------------|---------------------|------------|--------------|
| Initial-dropout label                   | 23 / 120            | 19.2       | 13.1 to 27.1 |
| Returned label among initial dropouts   | 15 / 23             | 65.2       | 44.9 to 81.2 |
| Non-return label among initial dropouts | 8 / 23              | 34.8       | 18.8 to 55.1 |
| Non-return label in the full summary    | 8 / 120             | 6.7        | 3.4 to 12.6  |
| Complement of the initial-dropout count | 97 / 120            | 80.8       | 72.9 to 86.9 |

Intervals are conditional on the compiled counts and a binomial independence assumption; they do not account for selection, distributor clustering or uncertain classification. The complement of initial dropout does not establish protocol completion.

Initial-dropout classifications increased across the age bands: 0 of 38 (0.0%), 7 of 41 (17.1%) and 16 of 41 (39.0%). The three-band chi-square test yielded  $\chi^2(2)=19.56$  ( $P<.001$ ), and the youngest-versus-oldest Fisher exact test also yielded  $P<.001$ . These are exploratory, unadjusted associations between age band and compiled labels. Among initial-dropout records, five of seven in the 45–64 band and ten of sixteen in the 65–80 band also carried a returned classification. The youngest band had no initial-dropout records, so its conditional returned percentage is undefined (Table 2; Figures 2 and 3).

Table 2. Age distribution of the compiled classifications.

| Age band in years | Records, n | Initial dropout, n (%) | Returned, n / dropout n | Non-return, n (%) |
|-------------------|------------|------------------------|-------------------------|-------------------|
| 25-44             | 38         | 0 (0.0)                | Not applicable          | 0 (0.0)           |
| 45-64             | 41         | 7 (17.1)               | 5 / 7                   | 2 (4.9)           |
| 65-80             | 41         | 16 (39.0)              | 10 / 16                 | 6 (14.6)          |
| Total             | 120        | 23 (19.2)              | 15 / 23                 | 8 (6.7)           |

Percentages in the initial-dropout and non-return columns use the age-band denominator. Returned classifications are a subset of initial dropouts. For ages 65–80, 10/16 returned classifications correspond to 62.5% (95% CI 38.6%–81.5%); 6/41 non-return classifications correspond to 14.6% (95% CI 6.9%–28.4%). The labels do not establish an observation-window endpoint or unprompted return.

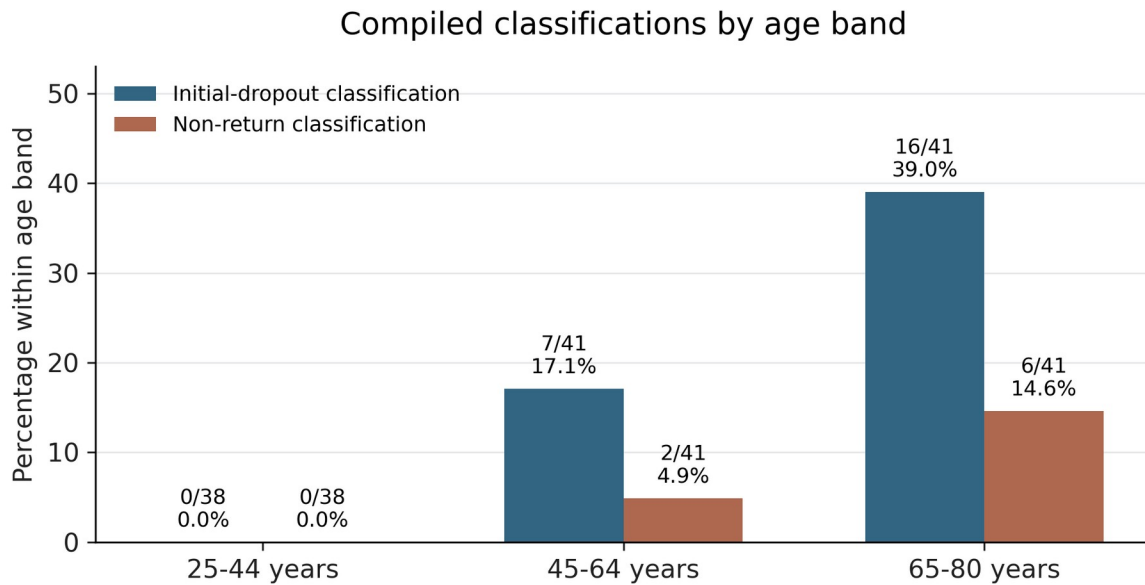


Figure 2. Initial-dropout and non-return classifications as percentages of each age band. Non-return classifications are a subset of initial-dropout classifications. The labels do not establish a defined end-of-window outcome.

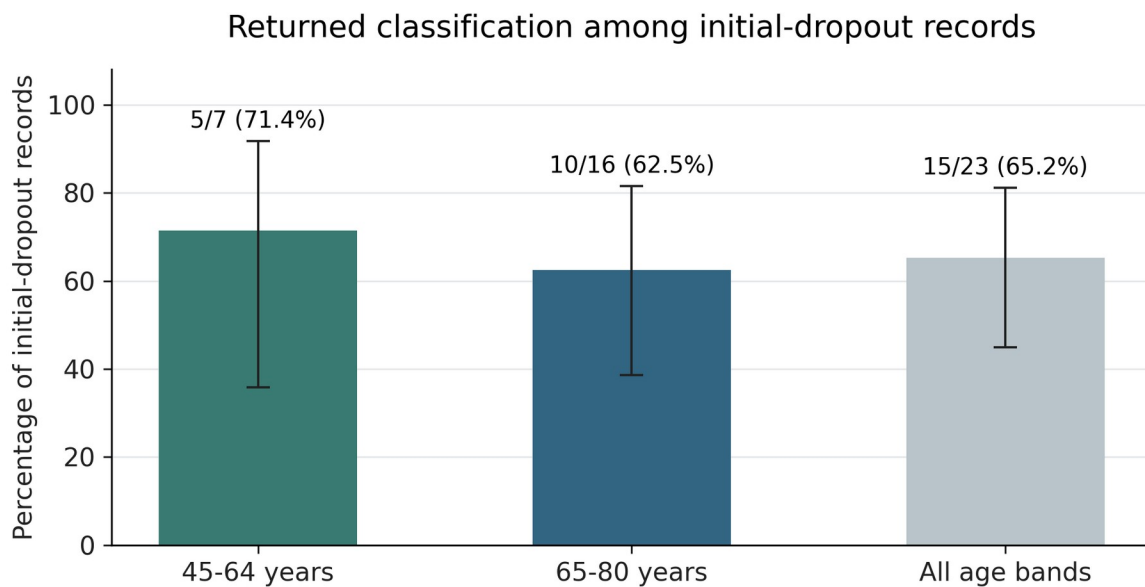


Figure 3. Returned classifications among initial-dropout records. Error bars are conditional Wilson 95% intervals from the reported counts. The 25–44-year band had no initial-dropout records, so its conditional percentage is undefined. Stage 2 is not plotted as a baseline or control.

## Questionnaire Coding in the Aggregate Summary

The summary records 86 of 120 responses (71.7%; 95% CI 63.0%–79.0%) as positive on the coordination-related item and 76 of 120 (63.3%; 95% CI 54.4%–71.4%) as positive on the regular-use-related item. The latter counts are 25 of 38, 28 of 41 and 23 of 41 across ascending age bands (Table 3). These are coding frequencies pending reconciliation with the administered instrument. The coordination-item age breakdown is unavailable. The summary provides no item-level missing-data flags. No “commitment gap” is inferred because the behavioural classifications and questionnaire items do not establish comparable or paired constructs.

Table 3. Positive-coded questionnaire responses reported in the aggregate summary.

| Summary item              | 25-44 years   | 45-64 years   | 65-80 years   | All records    |
|---------------------------|---------------|---------------|---------------|----------------|
| Coordination-related item | Not available | Not available | Not available | 86/120 (71.7%) |
| Regular-use-related item  | 25/38 (65.8%) | 28/41 (68.3%) | 23/41 (56.1%) | 76/120 (63.3%) |

Item labels are descriptive identifiers for the compiled counts. They do not replace the original administered wording, response scale or scoring threshold.

## Stage 3B Pro Developmental Findings

After reconciliation, 62 run appearances across the eight exports represented 34 distinct runs in 11 software-defined visits under four codes: 14 practice runs, six warm-ups and 14 performance checks. Repeated run records showed no conflicts. All runs recorded touch input. Three codes carried a prior-play response of yes and one of no; these responses do not identify a paired follow-up cohort from Stage 3A. Software visits do not necessarily represent separate days or independent players.

Eight decisions could be linked to the preceding performance and player-feedback records and to the settings recorded for the next run. All used local rules, with no cloud-model request. Fourteen feedback records comprised eight answered and six explicitly skipped responses. This demonstrates recorded implementation of feedback-informed adaptation, without establishing a benefit of adaptation or of a language model.

Three complete opening–closing performance-check pairs occurred under two codes. Obstacle clearance changed from 100% to 90%, 80% to 90%, and 80% to 90%, respectively (–10, +10 and +10 percentage points). Each check involved 20 obstacles with a nominal 30-second duration, but course versions differed and equivalence was not established. These mixed immediate changes, without a comparator or delayed unassisted assessment, do not demonstrate durable learning.

## Discussion

### Principal Findings

This study places self-appraisal at the centre of a question raised by abandonment of a clinically attached coordination game. Stage 3A implemented one way to investigate that question: make the game available outside the FRA apparatus and remove clinical outputs while retaining coordination-oriented play. The summary contained returned classifications for 15 of 23 initial-dropout records, including 10 of 16 in the oldest band, alongside an age gradient in initial-dropout labels. These observations motivate further investigation of how players interpret and revisit challenging experiences. They do not demonstrate a psychological mechanism or an observed disengagement–return sequence.

### Comparison With Prior Work

The comparison with the earlier clinical deployment is contextual. Stage 3A involved different participants, a wider age range, browser access, variable settings and assistance, and different equipment. Foot-pad and/or keypad control was intended to retain aspects of the earlier interaction, but does not make the deployments equivalent. Stage 3A players were new to this study game, whereas the Stage 2 residents had experience of the clinically attached system. Health-related

framing at recruitment is incompletely documented. Differences between these deployments cannot identify an isolated effect of decoupling.

FIAS proposes that a personally adverse interpretation of an assessment experience contributes to subsequent withdrawal. This is a hypothesis about self-appraisal, rather than a diagnosis inferred from low use. A useful test would measure perceived failure or capability threat after comparable experiences and link those measures prospectively to a defined return outcome. The Stage 3A zero-failure mode, absence of direct appraisal measures and unavailable event chronology prevent such a test. A bump is not necessarily experienced as failure. The earlier observation that three of six residents in the strongest-abandonment cluster showed a positive recorded balance-index change without returning [2] illustrates why measured performance and willingness to continue may diverge; it does not establish the cause of their withdrawal.

Existing acceptance and appraisal-based design frameworks already recognise self-efficacy, anxiety, meaning and personal interpretation [4,9,10]. The proposed contribution is therefore a specific, testable account of abandonment in the FRA context and a decoupling strategy for investigating it. The present design cannot exclude usability, game mechanics, access, fatigue or social support as contributing explanations. Age-band differences also do not establish uniform preferences or capabilities among older adults.

## **Game Achievement and Learning From Experience**

Three outcomes require distinct interpretation. Game achievement concerns completion, distance, points or other performance within a game. Clinical FRA performance requires the relevant assessment apparatus and a validated relation between its measurements and the clinical construct. Learning from experience concerns changes in skill, strategy or appraisal arising through practice and feedback. Stage 3A retained game achievement recording but provided no corresponding clinical FRA measurement; this report analyses aggregate engagement classifications rather than achievement trajectories. Removing clinical measurement changes what can be inferred from the recorded game data.

Continued participation creates an opportunity for learning but is not evidence that learning occurred. Immediate improvement may reflect familiarity, assistance or a change in difficulty. Durable learning requires suitable retention or transfer assessments [13]. Self-efficacy and stereotype-threat literature can guide hypotheses about how feedback is interpreted [14,15,16], but these mechanisms were not measured here. If clinically valid risk information is reintroduced, it should be communicated accurately with appropriate support. Game scores should not be presented as a diagnosis or a judgement of personal worth.

## **Stage 3B and Adaptive Learning Development**

Stage 3B was a subsequent exploratory development phase intended to examine whether alternative mechanisms could support adaptation and the measurement of learning from gameplay. The design retained a similar skiing aesthetic while changing mechanisms with assistance from Anthropic's Claude and introducing touchscreen input. The original component protocol, dated 8–9 August 2026, contrasted adaptive and static presentation and zero-failure and game-over modes; it did not manipulate clinical framing. Using Claude to help develop the software does not establish that a language model adapted play in real time.

Some Stage 3B players had previously played the study game, and its instructions asked about previous play. The original component phase yielded insufficient comparable data at the 26 September 2026 cut-off, including exclusions of returns on a superseded build. The September Pro exports are a separate developmental record: they demonstrate recorded operation of a rule-based adaptation process, without testing its learning benefit. Five of 11 software visits contained no practice, and three of the six visits with practice lacked a closing check. Prior exposure, input modality, changing builds and unresolved player attribution limit interpretation. Neither the original phase nor the Pro records are pooled with Stage 3A.

## **Limitations**

The unknown invitation denominator and nonprobability recruitment limit generalisation beyond the returned-record sample. The target of 200 does not establish how many people were approached or explain why files were not returned. Selecting only file-returning players may favour those willing and able to engage; the direction and magnitude of the overall bias cannot be estimated from the available record. A source-linked participant reconstruction is needed to verify unique participants, exclusions and variable-level missingness. Uncertain applied classification rules and event timing limit the behavioural meaning of the counts. Questionnaire wording and coding also require reconciliation.

Device combinations, assistance and distributor effects could influence participation but cannot be linked to individual outcomes. Changes across stages involve input modality, setting, mechanisms and prior experience, precluding attribution to any one change. No clinical outcome, identity-threat mechanism, durable learning outcome or language-model benefit was measured. The formal ethics decision followed Stage 3A collection; the chronology and the institution's determination remain material to submission and reporting.

## **Future Evaluation**

A test of the FIAS hypothesis could compare clinical and non-clinical presentations within the same recruitment population while holding equipment, mechanics and assistance constant. It should record exposure to framing, measure appraisal of capability and perceived failure, and prospectively define disengagement, prompting and a return window. Measures of usability, access and technology acceptance would help assess competing or interacting explanations. Prior play should be recorded and accounted for, with version identifiers, participant links and timestamps preserved. If fear of falling is relevant, a suitable instrument such as the Falls Efficacy Scale-International [17] could be included; it would not substitute for measuring the proposed self-appraisal mechanism and was not administered here.

A future adaptive-learning study should specify the target skill and what the adaptation changes, compare appropriate conditions using equivalent assessments, and measure delayed unassisted retention or transfer. Repeated participation by the same person must be identifiable and analysed accordingly, with separate identifiers for invited players. Game achievement, perceived improvement and durable learning should remain separate outcomes. The Pro logs support the technical implementation of feedback-informed adaptation; feasibility of a complete learning evaluation and its efficacy remain unestablished. Any further collection requires the applicable prospective ethics determination and prespecified hypotheses and endpoints.

## Conclusions

Decoupling a coordination game from FRA apparatus and outputs provides a design approach for investigating whether self-appraisal contributes to abandonment. The Stage 3A aggregate record describes age-related engagement classifications, but cannot establish FIAS, verify an effect of decoupling or demonstrate learning. Stage 3B extends the research direction toward adaptive learning with similar aesthetics and different mechanisms. Its Pro logs document rule-based adaptation, while evidence of learning remains inconclusive. The contribution is an exploratory account linking the appraisal question to a concrete design change and identifying the behavioural and learning measurements needed to test it.

## Acknowledgments

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## Funding

This research received no external funding.

## Data Availability

The aggregate Stage 3A data logbook and archived game build are retained with the first author's thesis records. The team retains returned exports and distribution and return correspondence recording implied consent. Participant-level compilation linked to the Stage 3A aggregates is pending. The eight Stage 3B Pro source exports are retained unchanged; their software metadata and the investigator's clarification of volunteer identities are distinguished in reporting. Requests for access should be directed to the corresponding author and require review of de-identification, participant consent and institutional conditions. A fully reconciled participant-level dataset has not been released.

## Author Contributions

SKTT led the research programme, deployment, compilation of records and manuscript preparation. CFC supervised the research programme and reviewed the preceding manuscript version.

## Conflicts of Interest

The authors declare no competing financial interests. The Stage-2 study was conducted with the residential care provider operating that facility; the provider had no role in the present Stage-3 study. Participants were recruited through the authors' networks.

## Use of Generative Artificial Intelligence

Anthropic's Claude Code assisted system development, including the HTML port and subsequent mechanism development. OpenAI's ChatGPT and Anthropic's Claude assisted manuscript revision, consistency review and plotting-code preparation for the 27 September 2026 version; ChatGPT assisted the revision and descriptive review of Pro records on 29 September 2026. Figures 2 and 3 were generated deterministically from the reported aggregate counts using plotting software. No participant observations were generated. The authors retain responsibility for the accuracy of the text,

references, analyses and figures. Development or writing assistance does not establish operation or effectiveness of a language model during gameplay.

## Related Publications

The programme's Stage 1 conference manuscript [1] and Stage 2 companion manuscript [2] concern separate study records. Their results are cited as background and are not new Stage 3A findings. The primary quantitative results use the March–April 2026 Stage 3A summary. September Stage 3B Pro logs are reported separately as developmental evidence and are not pooled with Stage 3A or the original Stage 3B component-phase records.

## Abbreviations

API: application programming interface

CI: confidence interval

FIAS: failure-induced abandonment syndrome

FRA: fall risk assessment

HTML: Hypertext Markup Language

STROBE: Strengthening the Reporting of Observational Studies in Epidemiology

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